Unit: 6 Unit 7: Factoring
Topic: Finding Common Factors

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the greatest common factor of 12, 18, and 24?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 24

Q2. Factor out the greatest common factor from $9x + 27$
(A) $3x + 9$
(B) $9(x + 3)$
(C) $3(3x + 9)$
(D) $27(x + 1)$
(E) $9(x + 3)$

Q3. What is the simplified expression for $8x^2 + 16x$ after factoring out the greatest common factor?
(A) $8x(x + 2)$
(B) $8(x^2 + 2x)$
(C) $16(x^2 + x)$
(D) $4x(2x + 4)$
(E) $2x(4x + 8)$

Q4. Factor $12x^3 + 18x^2$
(A) $6x^2(2x + 3)$
(B) $6x(2x^2 + 3x)$
(C) $3x(4x^2 + 6x)$
(D) $2x(6x^2 + 9x)$
(E) $6x^2(2x + 3)$

Q5. Simplify $20x + 16$ by factoring out the greatest common factor
(A) $4(5x + 4)$
(B) $4(4x + 5)$
(C) $5(4x + 4)$
(D) $4(4x + 4)$
(E) $20(x + 1)$

Q6. What is the greatest common factor of 15, 21, and 30?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 15

Q7. Simplify $32x^2y + 48xy$
(A) $16xy(2x + 3)$
(B) $8xy(4x + 6)$
(C) $16x(2xy + 3y)$
(D) $8x(4xy + 6y)$
(E) $16xy(2x + 3)$

Q8. Factor out the greatest common factor of $50x^3 + 30x^2$
(A) $10x^2(5x + 3)$
(B) $10x(5x^2 + 3x)$
(C) $5x^2(10x + 6)$
(D) $5x(10x^2 + 6x)$
(E) $10x^2(5x + 3)$

Open Ended Questions (FRQ)

Q9. Simplify $24x^3y + 36x^2y^2 + 48x^2y$ using factoring

Q10. Factor out the greatest common factor of $18x^2y + 12xy^2 + 6xy$ and simplify the expression

Chef's Tips

- Tip 1: Encourage students to use factoring for simplification
- Tip 2: Emphasize finding greatest common factors
- Tip 3: Allow students to use any method to factor the given expressions